

Aeroméxico Flight 498

Flight model: McDonnell Douglas DC-9

Date: 31 August ,1986

From: Mexico city, Mexico

To: Los Angeles, California

The ATC controller in LA is relatively new to the profession but had no significance to the below incident. A small airport is located near the LA airport which handles small charter flights and piper planes.

Problem

During Day, the pilots finally prepare to land the plane and enter the **terminal control area (TCA)**, TCA is a 3 dimensional airspace in and around the airport. At the same time a piper plane also enters the TCA. The DC9 experiences a sudden shock and starts to loss altitude rapid. It then crashes into a residential building. All passengers boarded the DC9 are confirmed dead. The plane had damages almost 16 houses in the residential neighborhood.

Investigation

In the nearby the crash site, the investigators find a piper crashed. The plane canopy is found to be slides off. The propeller marks on the DC9 paints how the 2 plane could have crashed. It is found the the piper had collided with the tail of the DC9 and the horizontal stabilizer has broken off. A plane with broken horizontal stabilizer losses the pitch control. On analyzing the flight data recorder, it is found that the DC9 was well within the TCA as informed by the controller. The piper was also inside the TCA. For a plane to enter the TCA, it must have got prior permission from the controller. It was found that the flight intended plan for the piper was different that the path that they had taken, the intended path avoid the TCA. It is found that the piper's pilot under control may have suffered a heart attack, which was ruled out later upon further analysis. The ATC controller says that he failed to see any piper plane on his radar. A possible answer for the piper to enter into the TCA is that the plane fails to have complex navigation systems and has to be navigated using visual clues which could have miss guided the piper jet. The controllers are positive that the piper was missing on radar but the investigators are convinced that the controllers must be able to see the plane. A possible explanation to the same is that when a beam of radar is sent to the plane, the plane responds back by sending another beam of signals and if this response beam of signal gets blocked by the any building then the would disappear from radar. But the piper lack the transponder necessary for the ATC to get information regarding the plane on the height and speed.

Changes that were brought after the incident

The FAA instructed all the small planes to be fitted with a transponder which would send all the required information about the plane to the near by ATC. Also the ATC computes are programmed with a new program which detects any intruders into the airspace. This program is called **Mode C Intruder alert**. Also this incident made it mandate that all plane flying should have TCAS fitted in their plane.

Reference

1. A Labour Day Weekend SHATTERED By A Devastating Tragedy — Flight 498 — May-day: Air Disaster
<https://www.youtube.com/watch?v=oFbssXW0IJE>